SE LAB 3

Component Modelling & Architectural Pattern Selection

Problem Statement 06

Alumni Mentorship & Mock Interview Platform

Name: Daruru Bhuvan Sai
SRN: PES1UG24AM356
Section: F

We chose Microservices Architecture for the Alumni Mentorship & Mock Interview Platform System.

Reason 1 — Independent Scaling Under Peak Load

NFR-002 requires mentor recommendations and availability to return within 2 seconds for 95% of requests under 500 concurrent users. Only the Mentor Recommendation and Booking & Scheduling services face this search/booking peak, so they can be scaled independently (extra instances, caching) without over-provisioning lower-traffic services such as Scorecard & Feedback.

Reason 2 — Fault Isolation Across Independent Workflows

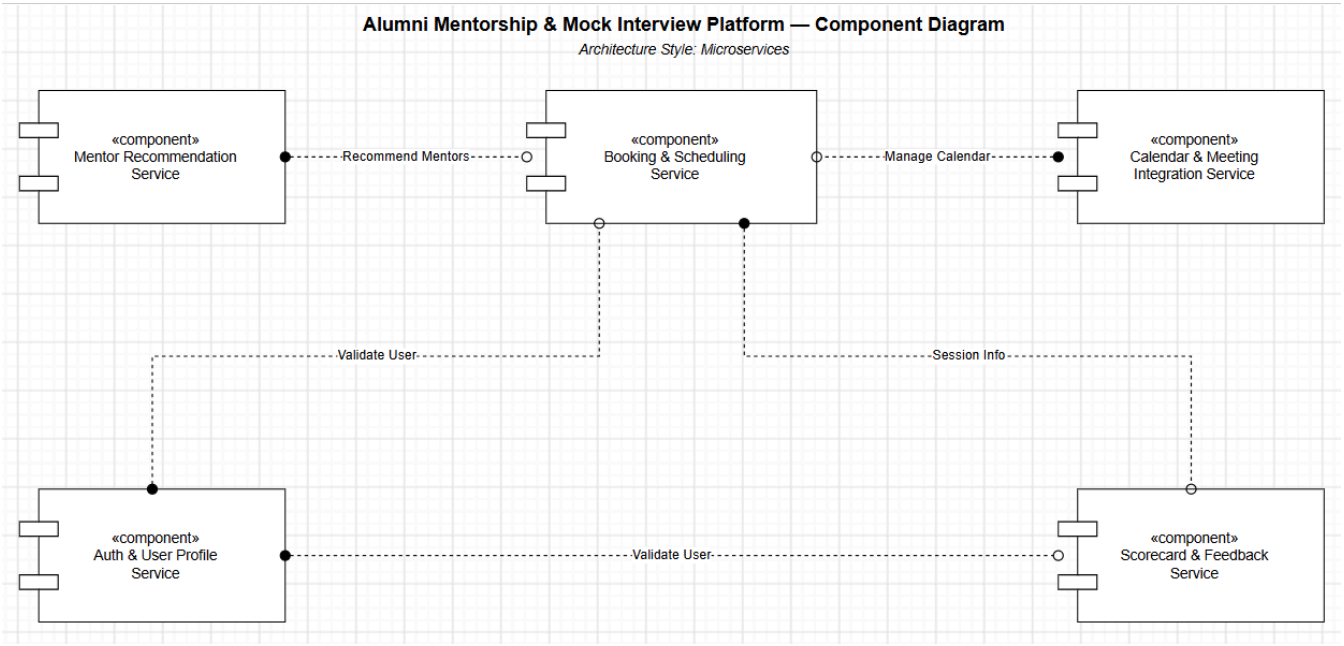
The platform depends on an external Calendar & Meeting Integration service to create meeting links (FR-001, FR-003). If this integration is slow or temporarily unavailable, students can still browse recommended mentors and mentors can still manage their slots, because each capability runs as a separate deployable service rather than one tightly coupled layer.

Security Advantage

NFR-001 requires that user profiles and feedback scorecards be protected, with soft-deletion of a user's data within 24 hours of a valid request. Isolating the Scorecard & Feedback service from the other services lets it enforce its own access-control and audit-logging boundary for this sensitive academic feedback. Every cross-service call is routed through the Auth & User Profile service's ValidateUser interface, so a student can never reach another student's scorecard, and the soft-delete of a user's data can be applied within that one service without touching unrelated services.

Performance Benefit

Because the Mentor Recommendation service is deployed independently of booking, calendar, and scorecard logic, it can be replicated and cached to absorb the 500-concurrent-user peak specified in NFR-002 while keeping response time under 2 seconds, without adding cost or latency to the less frequently used scorecard and calendar workflows.


Alumni Mentorship & Mock Interview Platform — Component Diagram
Architecture Style: Microservices
«component»
Mentor Recommendation
Service
Recommend Mentors
«component»
Booking & Scheduling
Service
Manage Calendar
«component»
Calendar & Meeting
Integration Service
Validate User
Session Info
«component»
Auth & User Profile
Service
Validate User
«component»
Scorecard & Feedback
Service